Arael A. Anaya

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EDUCATION

Colorado School of Mines | Golden, CO — **May 2027**
Master of Science in Robotics Engineering — GPA: 4.0
Bachelor of Science in Mechanical Engineering; Focus: Automation & Control — GPA: 3.7
Minor: Robotics and Intelligent Systems

WORK EXPERIENCE

Navflex | Robotics Engineer Intern | Broomfield, Colorado — **May 2026 - August 2026**

- Designed a full replacement of the vehicle control system, transitioning from a static PID architecture to an adaptive controller that self-tuned in real time to optimize performance across operating conditions.
- Reverse-engineered and mapped a large-scale ROS codebase end-to-end, developing a deep understanding of the full autonomy pipeline spanning perception, localization, planning, and controls.
- Identified control bottlenecks and proposed targeted algorithmic improvements through systematic analysis of live system behavior and logged data.
- Collaborated with a team of 30 engineers in daily scrum workflows, contributing production-ready code and participating in structured code reviews.
- Leveraged Docker for containerized development and deployment, maintaining consistent environments across the engineering team.

MPALA Lab | Research Assistant | Colorado School of Mines — **Jan 2025 - Present**

- Developed a ROS2-integrated aerial swarm pipeline using Crazyflie drones and OptiTrack motion capture to study distributed consensus algorithms in real-world networks.
- Implemented resilient consensus algorithms (W-MSR), tuned Kalman filters, and system dynamics to improve stability, localization accuracy, and robustness under adversarial interference.
- Bridged theory and practice by translating consensus models into real-time experiments, conducting flight tests, and performing real-time debugging.
- Collaborated in agile development workflows, contributing high-quality C++ & Python code to the research team.

RELEVANT PROJECT EXPERIENCE

Crank Generator (Human-Powered Device) — **Aug 2025 - Dec 2025**

- Designed and prototyped a custom human-powered generator, performing SolidWorks modeling, FEA validation, and design-for-manufacturing optimization.
- Led fabrication and testing efforts, validating structural integrity and real-world performance.

Quadcopter Altitude Control System — **Aug 2025 - Dec 2025**

- Designed and simulated a closed-loop altitude PID controller in MATLAB/Simulink, analyzing step and sinusoidal responses.
- Evaluated rise time, settling time, overshoot, and steady-state error; improved disturbance rejection and control performance by approximately 20%.
- Developed a repeatable gain-tuning and validation workflow, including automated testing scripts and structured code reviews through GIT.

Rescue Drone — **Aug 2024 - Dec 2025**

- Designed and built an entirely custom quadcopter system with claw mechanism for S-A-R payload delivery.
- Integrated multiple microcontrollers in a distributed embedded architecture, programming real-time firmware.
- Performed full mechanical, electrical, and software integration, including CAD, additive manufacturing, soldering, and flight testing.

ENGINEERING AND TECHNICAL SKILL

- **Programming**: Python, C++, C, MATLAB, Java, JavaSwing, JavaScript, LabVIEW, EES, HTML, CSS
- **Tools/Platforms**: ROS2, Linux, Git, Docker, Agile development, CMake, Powershell, bash, Pytests
- **Hardware & Electronics**: Microcontroller Programming, Electrical Soldering, Additive Manufacturing, Lidar
- **CAD & Fabrication**: SolidWorks (CSWP, Advanced, CSWP-S, and MBD Certified), CNC, Drill Press, Lathe
- **Soft Skills**: Fluent in English and Spanish, Team Leadership, Project Management, Effective communication

AWARDS AND PUBLICATIONS

- SPIRSE Travel Grant - $3,000 award by IEEE to attend CASE - Recognized for robotics research potential — **2025**
- SURF 2025 Scholar - Summer Undergraduate Research Fellowship — **2025**
- MURF 2026 Scholar - Undergraduate Research Fellowship — **2025**
- [PENDING PAPER] Understanding Human Operator Needs in Multi-Robot Tasking Interfaces — **2025**